

Project Name	Import Data using Transform Maps	Maximum Marks	2 Marks
Date	31 January 2025	Team ID	6ab615fddcdae33acb38af6f

Proposed Solution Details

The team's proposed solution is described below by parameter: the problem it solves, the idea, what makes it new, its impact on people, the business model, and how it scales.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Organizations face operational inefficiency, data corruption, and compliance risk when importing and managing bulk employee training records. Manual CSV uploads lead to: • duplicate records; • broken foreign-key relationships, such as manager and department references; • data exposure caused by inadequate field-level security controls.
2	Idea / Solution Description	An enterprise ServiceNow data ingestion and transformation pipeline built on Import Sets and custom Transform Maps. It combines: • automated coalesce keys (u_emp_id) for intelligent deduplication; • custom onBefore transform scripts for data validation; • dynamic dot-walking that resolves manager and department references against the sys_user and cmn_department tables; • role-based Access Control Lists (ACLs) for secure access.
3	Novelty / Uniqueness	Combines automated multi-source data ingestion, real-time dynamic dot-walking reference resolution, and ACL-backed security in one unified ServiceNow application. It removes manual spreadsheet cleansing, automates relationship mapping, and enforces strict compliance without external middleware .
4	Social Impact / Customer Satisfaction	• Reduces administrative fatigue and data entry errors for HR professionals. • Gives employees transparent, secure access to their training transcripts and certifications. • Gives leadership and compliance auditors reliable, auditable reporting.
5	Business Model (Revenue Model)	An enterprise SaaS model built on internal operational efficiency. It can be deployed as a certified ServiceNow Store solution with tiered licensing based on active employee user seats and automated compliance auditing volume.
6	Scalability of the Solution	Built on ServiceNow's cloud-native, multi-tenant architecture. Asynchronous batch import sets, scheduled background data transformers, and indexed database fields let the solution scale to large volumes of employee training records across global business units while keeping performance steady.

Solution at a Glance

The four building blocks of the solution, each tied to a problem it removes.

Feature	What It Does	Problem Removed
Coalesce keys	Deduplicate on the employee key	Duplicate records
onBefore scripts	Validate rows before insert	Corrupted or invalid data
Dot-walking	Resolve manager and department	Broken references
Role-based ACLs	Restrict access to scores	Data exposure

Success Metrics and Ownership

Suggested measures of success and the role best placed to own each one.

Metric	Target	Suggested Owner
Duplicate records after import	0 when the same file is imported twice	System Administrator
Reference accuracy	All rows resolve employee, manager, and department	System Administrator
Access control	Unauthorized users cannot read or edit training scores	HR Compliance Lead
Manual cleanup effort	No hand-fixing of records after import	HR Operations

Metrics and owners are suggestions for the team to confirm.

Risks and Mitigations

Risk	Mitigation
Missing or non-standard employee IDs in source files	Validate IDs before import and reject blank coalesce keys.
Incomplete manager or department data in sys_user	Check user data completeness first and flag empty dot-walked values.
Over- or under-restrictive ACLs	Test with both an authorized and an unauthorized user before go-live.
Slow imports on very large files	Use asynchronous batch imports and index the coalesce fields.